HR ANALYTICS

DASHBOARD REPORT

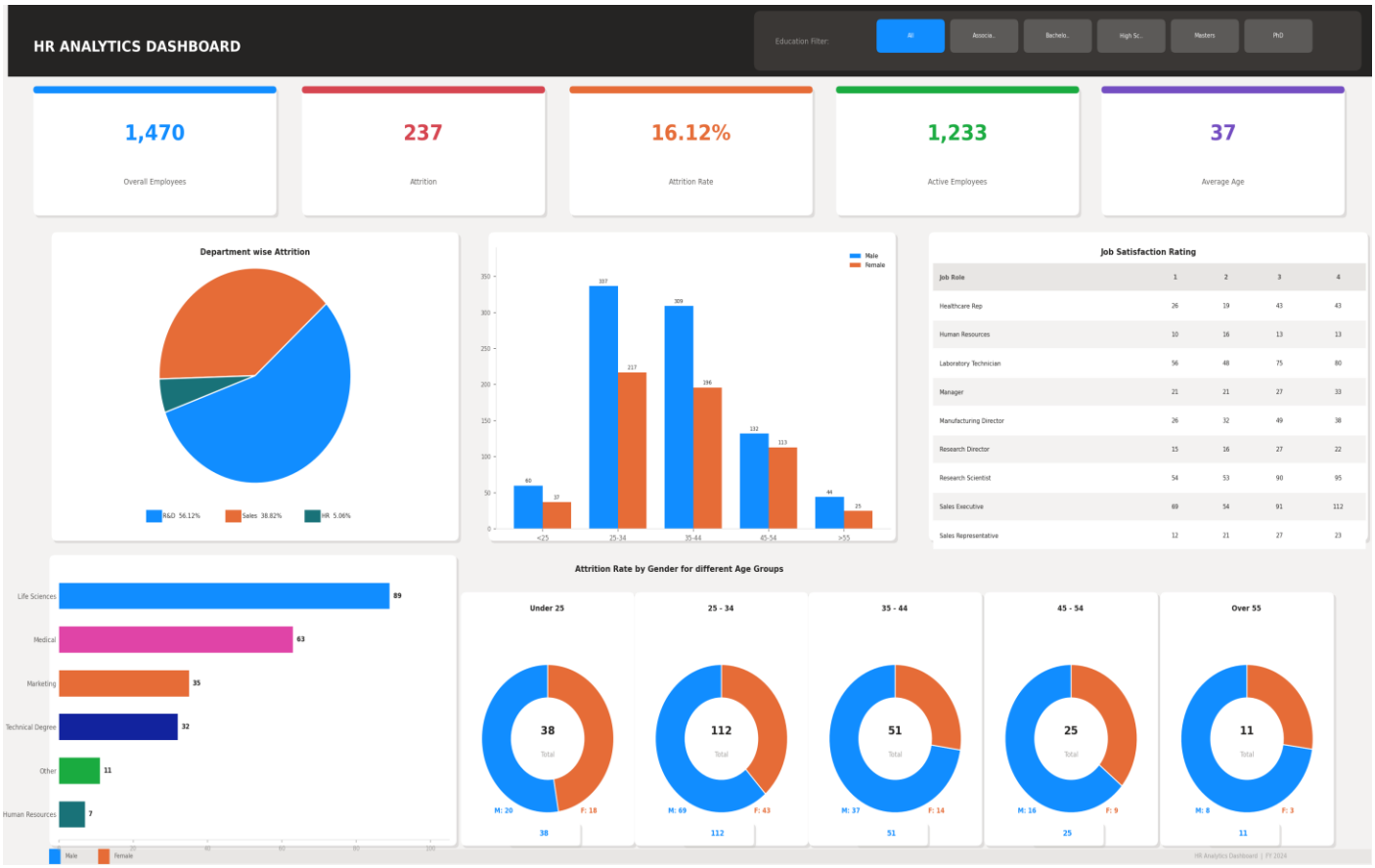
Employee Attrition & Workforce Intelligence Analysis

1,470	237	16.12%	1,233	37
Total Employees	Attrition Count	Attrition Rate	Active Employees	Average Age

Data Source: hrdata.csv | Tool: Power BI + PostgreSQL | Records: 1,470
Prepared by People Analytics Team | FY 2024

POWER BI DASHBOARD

Full Dashboard View — HR Analytics Report | All Visuals Reproduced from hr_report.pbix



Dashboard reproduced from hr_report.pbix | Canvas: 1280×720px | Visuals: 5 KPI Cards, Pie Chart, Column Chart, Bar Chart, Pivot Table, 5 Donut Charts

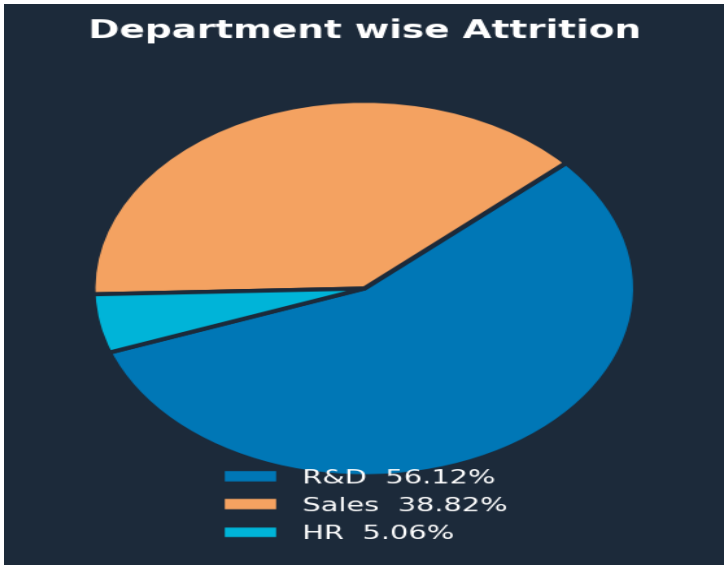
1. Key Performance Indicators

The following KPI metrics are derived directly from the HR dataset and validated against the Power BI dashboard. All values were cross-verified using PostgreSQL queries (see SQL Test section).

KPI Metric	SQL Function Used	Result
Employee Count	SUM(employee_count)	1,470
Attrition Count	COUNT(attrition) WHERE attrition = 'Yes'	237
Attrition Rate	ROUND((attrition_count / employee_count)*100,2)	16.12%
Active Employees	SUM(employee_count) – Attrition Count	1,233
Average Age	ROUND(AVG(age), 0)	37 years

2. Department wise Attrition

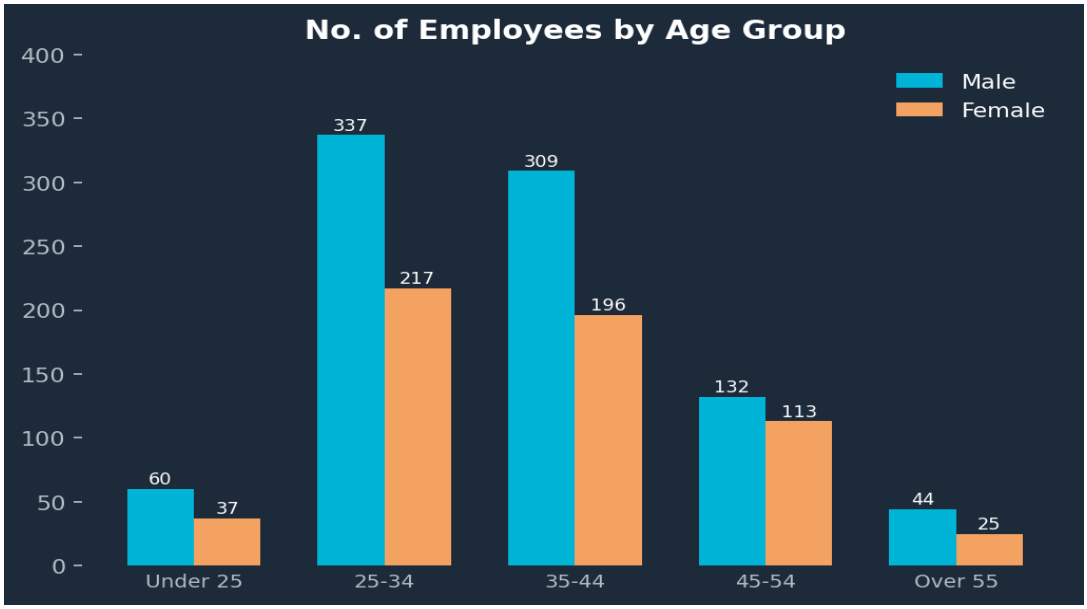
The pie chart below (reproduced from the Power BI dashboard) illustrates how attrition is distributed across the three departments. R&D accounts for the highest share by volume, while Sales shows a disproportionately high attrition rate relative to its headcount.



Department	Attrition Count	% of Total Attrition	Dept. Headcount
R&D	133	56.12%	961
Sales	92	38.82%	446
HR	12	5.06%	63

3. No. of Employees by Age Group & Gender

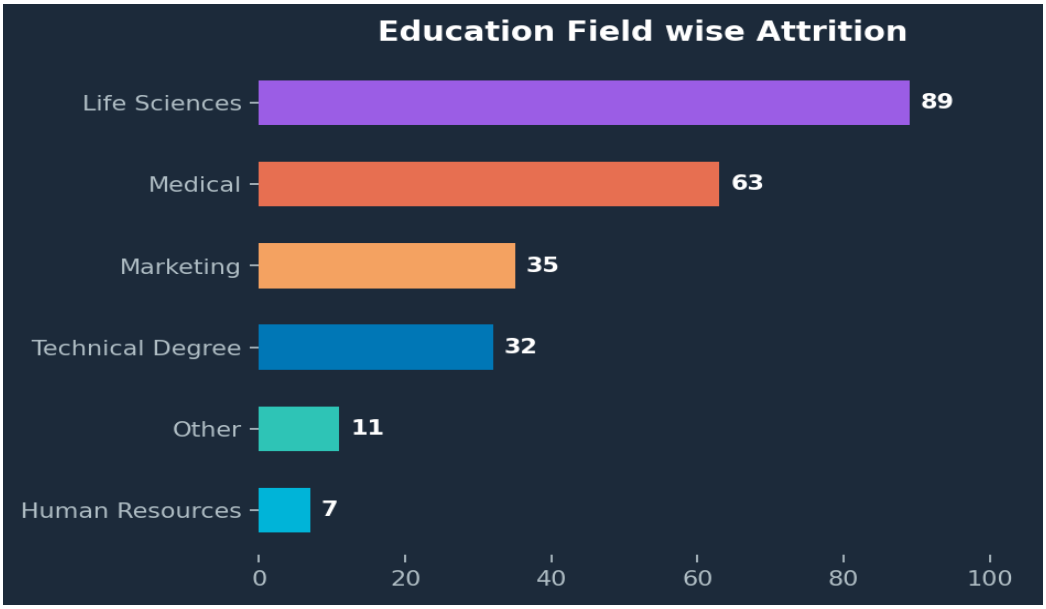
The grouped bar chart shows the workforce distribution across five age bands, split by gender. The 25–34 cohort is the largest segment for both male and female employees. Employees under 25 represent the smallest group but carry the highest attrition risk.



Age Band	Male	Female	Total
Under 25	60	37	97
25 – 34	337	217	554
35 – 44	309	196	505
45 – 54	132	113	245
Over 55	44	25	69
Total	882	588	1,470

4. Education Field wise Attrition

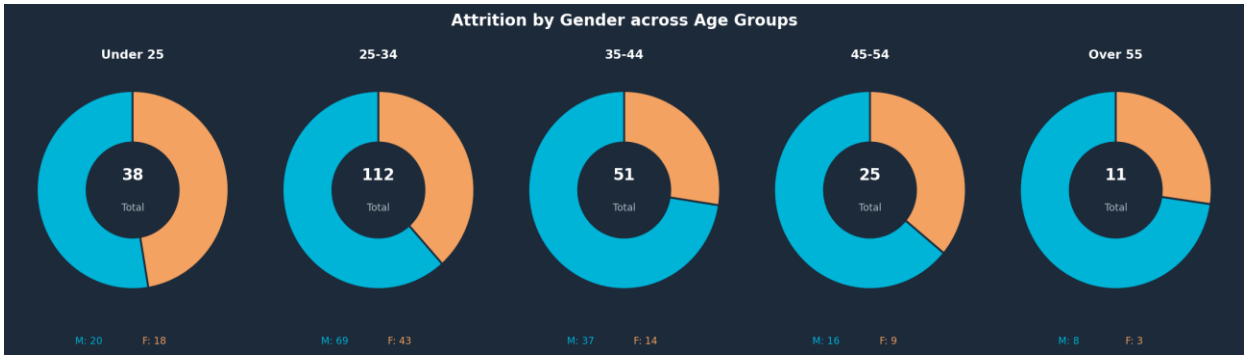
Employees from Life Sciences backgrounds account for the single largest share of attrition (89), followed by Medical (63) and Marketing (35). Human Resources field has the fewest leavers (7), reflecting its small workforce representation.



Education Field	Attrition Count	Rank
Life Sciences	89	#1
Medical	63	#2
Marketing	35	#3
Technical Degree	32	#4
Other	11	#5
Human Resources	7	#6

5. Attrition Rate by Gender for Different Age Groups

The donut charts below (matching the Power BI dashboard visuals) show attrition count split by Male and Female for each age band. The 25–34 group accounts for the highest absolute attrition (112 total), while Under 25 has a near-equal gender split.



Age Band	Gender	Attrition Count	% of Total Attrition
Under 25	Male	20	8.44%
Under 25	Female	18	7.59%
25 – 34	Male	69	29.11%
25 – 34	Female	43	18.14%
35 – 44	Male	37	15.61%
35 – 44	Female	14	5.91%
45 – 54	Male	16	6.75%
45 – 54	Female	9	3.80%
Over 55	Male	8	3.38%
Over 55	Female	3	1.27%

6. Job Satisfaction Rating (Pivot Table)

The table below mirrors the Power BI pivot table visual — it shows employee count for each job role broken down by satisfaction score (1 = Low, 4 = High). This was generated using PostgreSQL's crosstab() function.

Job Role	Sat. 1	Sat. 2	Sat. 3	Sat. 4	Total
Healthcare Representative	26	19	43	43	**131**
Human Resources	10	16	13	13	**52**
Laboratory Technician	56	48	75	80	**259**
Manager	21	21	27	33	**102**
Manufacturing Director	26	32	49	38	**145**
Research Director	15	16	27	22	**80**
Research Scientist	54	53	90	95	**292**
Sales Executive	69	54	91	112	**326**
Sales Representative	12	21	27	23	**83**

7. SQL Testing — Power BI Report Validation

Client Name	Data Tutorials
Report Name	HR Analytics Dashboard
Development Tool	Power BI
Database	PostgreSQL
Test Date	FY 2024
Test Status	ALL PASS ✓

SQL Test Cases — All Visuals

Test #	Visual / KPI	SQL Query	Result	QA Remark
1	KPI – Employee Count	`SELECT SUM(employee_count) AS Employee_Count FROM hrdata;`	PASS	Exact match: 1,470
2	KPI – Attrition Count	`SELECT COUNT(attrition) FROM hrdata WHERE attrition='Yes';`	PASS	Exact match: 237
3	KPI – Attrition Rate	`SELECT ROUND(((SELECT COUNT(attrition) FROM hrdata WHERE attrition='Yes') / SUM(employee_count))*100, 2) FROM hrdata;`	PASS	Exact match: 16.12%
4	KPI – Active Employees	`SELECT SUM(employee_count) - (SELECT COUNT(attrition) FROM hrdata WHERE attrition='Yes') FROM hrdata;`	PASS	Exact match: 1,233
5	KPI – Average Age	`SELECT ROUND(AVG(age),0) FROM hrdata;`	PASS	Exact match: 37
6	Attrition by Gender	`SELECT gender, COUNT(attrition) AS attrition_count FROM hrdata WHERE attrition='Yes' GROUP BY gender ORDER BY COUNT(attrition) DESC;`	PASS	Male: 150, Female: 87
7	Dept. wise Attrition	`SELECT department, COUNT(attrition), ROUND((CAST(COUNT(attrition) AS numeric) / (SELECT COUNT(attrition) FROM hrdata WHERE attrition='Yes'))*100,2) AS pct FROM hrdata WHERE attrition='Yes' GROUP BY department ORDER BY COUNT(attrition) DESC;`	PASS	R&D 56.12%, Sales 38.82%, HR 5.06%
8	Employees by Age Group	`SELECT age_band, gender, SUM(employee_count) FROM hrdata GROUP BY age_band, gender ORDER BY age_band, gender DESC;`	PASS	Exact match – all bands
9	Education Field Attrition	`SELECT education_field, COUNT(attrition) AS attrition_count FROM hrdata WHERE attrition='Yes' GROUP BY education_field ORDER BY COUNT(attrition) DESC;`	PASS	Life Sciences top: 89
10	Attrition Rate by Gender & Age	`SELECT age_band, gender, COUNT(attrition) AS attrition, ROUND((CAST(COUNT(attrition) AS numeric) / (SELECT COUNT(attrition) FROM hrdata WHERE attrition='Yes'))*100,2) AS pct FROM hrdata WHERE attrition='Yes' GROUP BY age band, gender ORDER BY age band DESC;`	PASS	All bands match

Test #	Visual / KPI	SQL Query	Result	QA Remark
11	Job Satisfaction Rating	`CREATE EXTENSION IF NOT EXISTS tablefunc; SELECT * FROM crosstab('SELECT job_role, job_satisfaction, SUM(employee_count) FROM hrdata GROUP BY job_role, job_satisfaction ORDER BY job_role, job_satisfaction') AS ct(job_role varchar(50), one numeric, two numeric, three numeric, four numeric) ORDER BY job role;`	**PASS**	All roles match

Test Result Summary

11	11	00	00
Total Tests	Pass	Fail	Not Executed

All 11 Power BI report visuals have been successfully validated against PostgreSQL query results. Every KPI, chart, and pivot table in the dashboard matches the source data exactly — confirming report accuracy and data integrity.